

# Story of Geometry

Part 1





## About the Course

This course gives students a glimpse into how the ideas that guide our study of geometry were first discovered.



## Placement & Combining Tips

This course is intended as a precursor to Part 2; however, the order may be reversed to facilitate combining students.



## Scheduling

| GRADE | SCHEDULE INFO.                                  | BOOKS                                                                                                                  |
|-------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| 7     | Story of Geometry: Part 1<br>1 time/week 15 min | String, Straight-Edge, and Shadow<br>The Student Bible Atlas, Revised Edition<br>Historical Atlas of the World 6th Ed. |

### Sample Weekly View

| Day 1                     | Day 2 | Day 3                        | Day 4 | Day 5 |
|---------------------------|-------|------------------------------|-------|-------|
| Story of Geometry: Part 1 |       |                              |       |       |
|                           |       | Story of Geometry:<br>Part 1 |       |       |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

### Story of Geometry: Part 1



String, Straight-Edge, and Shadow



The Student Bible Atlas, Revised Edition



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)  
(No Subject Supplies Assigned)



## Quick Links

(No Quick Links Assigned)

[Click THIS text  
or scan the QR  
code for links.](#)



# Story of Geometry: Part 1

## How To Teach



### Recap

Connect back to previous lesson.



### Introduce

Optional: Use a picture, map, or engaging question to prepare the student to receive the idea from the day's lesson. Talk about one or two unknown words that will be crucial to understanding.



### Read

Read the new passage out loud or have the student read it orally or silently.



### Narrate

Have the student narrate orally or in writing.



### Discuss

- Give students an opportunity to share their thoughts on the lesson and any connections made.



### Practice

Optional: Attempt to recreate some of the problems from the stories.

# Story of Geometry: Part 1

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## Term 1

### WEEK 1 □ 15m Story of Geometry: Part 1 - Lesson 1

#### *Introduction*

□ Materials: String, Straight-Edge, and Shadow

#### → INTRO

As you begin your study of geometry, this book will help you understand some concepts better by showing you how they were discovered and used to solve real problems.

#### → READ, NARRATE, & DISCUSS

"Prologue" p.7-10 (UK 7-9) "The string, the" - "and the shadow."

### WEEK 2 □ 15m Story of Geometry: Part 1 - Lesson 2

#### *The Natural Sense of Order*

□ Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.1 p.12-13 (UK 12-13) "Geometry itself" - "built into them."

- Look around the room in which you are sitting or outside. Write a paragraph telling what elements of mathematics (including geometry) you see.

### WEEK 3 □ 15m Story of Geometry: Part 1 - Lesson 3

#### *Mathematics as Beauty*

□ Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.1 p.13-14 (UK 13-14) "Within ourselves" - "geometry really began."

### WEEK 4 □ 15m Story of Geometry: Part 1 - Lesson 4

#### *From Fear to Wonder*

□ Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.2 p.15-16 (UK 15-16) "When and how did" - "an amazing surprise."

#### ★ TEACHER NOTE

You may wish to preview p.15, which talks about the age of the earth. Also, please note the illustration of the partially nude woman on p.15 (UK 15).

### WEEK 5 □ 15m Story of Geometry: Part 1 - Lesson 5

#### *Structure in Nature*

□ Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.2 p.17-18 (UK 16-17) "Before we learn" - "of living things."

# Story of Geometry: Part 1

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## Term 1

- How could seeing basic geometric shapes help you with nature study?
- Describe an object in terms of its shape. Use geometry words such as line, curve, spiral, perpendicular, etc.

### WEEK 6 15m Story of Geometry: Part 1 - Lesson 6

*Geometry of Living Things*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.2 p.18-19 (UK 17-18) "Did you ever" - "five-pointed star."

### WEEK 7 15m Story of Geometry: Part 1 - Lesson 7

*From the Art Gallery of the Universe*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.2 p.19-20 (UK 18-19) "Best of all" - "dim Stone Age."

### WEEK 8 15m Story of Geometry: Part 1 - Lesson 8

*Technics and Art*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.3 p.21-22 (UK 20-21) "Stone Age men" - "variety of nature's forms."

#### ★ TEACHER NOTE

Please note the illustration of the partially nude woman on p.23 (UK 21).

### WEEK 9 15m Story of Geometry: Part 1 - Lesson 9

*Design Mimics Nature*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.3 p.22-24 (UK 21-22) "Perhaps unconsciously" - "using circles."

#### ★ TEACHER NOTE

Please note the illustration of the partially nude woman on p.23 (UK 21).

### WEEK 10 15m Story of Geometry: Part 1 - Lesson 10

*Circles*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → INTRO

Where in this room do you see circles?

#### → READ, NARRATE, & DISCUSS

Ch.3 p.24-25 (UK 22) "Today we are" - "they gave up!"

# Story of Geometry: Part 1

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## Term 1

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div><div>WEEK 11</div><div><div><div></div></div>15m Story of Geometry: Part 1 - Lesson 11</div><div>Drawing a Perfect Circle with String</div><div><div></div>Materials: String, Straight-Edge, and Shadow</div><div><div>→ RECAP</div><div>What did you read last time?</div><div>→ INTRO</div><div>How would you define an "accurate" circle?</div><div>→ READ, NARRATE, &amp; DISCUSS</div><div>Ch.3 p.26-27 (UK 22-24) "So, you see" - "the Stone Age."</div></div></div> |  |
| <div><div>WEEK 12</div><div><div><div></div></div>15m Story of Geometry: Part 1 - Lesson 12</div><div>Exam Week</div><div><div>→ EXAMS</div><div>Answer question(s) related to course.</div><div>Questions will come from:</div><div>String, Straight-Edge, and Shadow</div></div></div>                                                                                                                                                                                        |  |

# Story of Geometry: Part 1

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## Term 2

### WEEK 1 15m Story of Geometry: Part 1 - Lesson 13

#### *Agrarian Society*

Materials: String, Straight-Edge, and Shadow; Student Bible Atlas

#### → INTRO

This section is titled "Reading the Shadows." What do you think that means?

#### → READ, NARRATE, & DISCUSS

Ch.4 p.29-30 (UK 26-27) "Here our story" - "of the shadows."

- This map shows the Fertile Crescent. What do you notice about the surrounding land?

Map Resource: Student Bible Atlas "Map 1: Lands of the Bible" p.4-5

### WEEK 2 15m Story of Geometry: Part 1 - Lesson 14

#### *Noticing Shadows*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.4 p.30-32 (UK 27-29) "Perhaps it was" - "a repeated pattern."

### WEEK 3 15m Story of Geometry: Part 1 - Lesson 15

#### *Measuring Time and Seasons*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.4 p.32-35 (UK 29-31) "The sun seemed" - "the fruitful year."

- Look at a sundial. How is it similar to, and different from, a clock?

∞ Image Link: Sundial

#### → OBSERVE

Choose one of the ways to observe shadows from the book to try yourself.

### WEEK 4 15m Story of Geometry: Part 1 - Lesson 16

#### *Usefulness of Right Angles*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → INTRO

A surveyor is someone who makes very precise measurements of land to determine property boundaries. Why do you think that might be important? Look at the picture of a modern surveyor. This story will show how this profession got its start.

∞ Image Link: Surveyor

#### • IMPORTANT DATES

**Village of Jarmo:** First Evidence of Straight Lines & Right Angles (c.5000 BC, Syria)



# Story of Geometry: Part 1

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## Term 2

### → READ, NARRATE, & DISCUSS

Ch.5 p.36-38 (UK 32-34) "Now at last" - "on the earth?"

#### WEEK 5 15m Story of Geometry: Part 1 - Lesson 17

*The 3-4-5 Right Triangle*

Materials: String, Straight-Edge, and Shadow

### → RECAP

What right angles did you notice last week?

### → READ, NARRATE, & DISCUSS

Ch.5 p.38-41 (UK 34-37) "We know part" - "perfect right angle."

### → PRACTICE

Using your geometry tools, try to construct a 3-4-5 right triangle.

### • IMPORTANT DATES

Rope Stretchers (First Surveyors; c.3000 BC, Egypt)

#### WEEK 6 15m Story of Geometry: Part 1 - Lesson 18

*The Plumb-Line*

Materials: String, Straight-Edge, and Shadow

### → RECAP

Tell about a perfect right triangle.

### → READ, NARRATE, & DISCUSS

Ch.5 p.42-44 (UK 37-39) "Now that surveyors" - "way to keep peace."

- Where are levels and plumb lines important today?

#### WEEK 7 15m Story of Geometry: Part 1 - Lesson 19

*Area and Architecture*

Materials: String, Straight-Edge, and Shadow

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

Ch.5 p.44-46 (UK 39-41) "Again the idea" - "of the temple."

### • IMPORTANT DATES

Oldest Man-Made Stone Structure (Step-Pyramid at Sakkara, c.2750 BC)  
Great Pyramid (c.2850 BC)

#### WEEK 8 15m Story of Geometry: Part 1 - Lesson 20

*Early Astronomy*

Materials: String, Straight-Edge, and Shadow

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

Ch.6 p.47-50 (UK 42-44) "Meantime on the" - "along the Nile."

#### WEEK 9 15m Story of Geometry: Part 1 - Lesson 21

*Arcs*

Materials: String, Straight-Edge, and Shadow

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

# Story of Geometry: Part 1

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## Term 2

Ch.6 p.50-52 (UK 44-46) "The Egyptians" - "exactly six equal parts."

### → PRACTICE

Using your geometry tools, try to construct a circle and divide it into six equal parts. Use the following pictures to help you.

📎 Image Resource: String, Straight-Edge, and Shadow p.52-53 (UK 46-47)

### WEEK 10 📅 15m Story of Geometry: Part 1 - Lesson 22

*Measuring Star Travel*

📎 Materials: String, Straight-Edge, and Shadow

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

Ch.6 p.52-54 (UK 46-48) "Anyhow, by chance" - "celestial bodies."

### → PRACTICE

Into how many equal parts is the circle of a clock divided by the hour marks? Using the same method you used last week to construct a circle with equal parts, construct a clock face (hours only).

### • IMPORTANT DATES

First Continuous Astronomical Data Started (747 BC, Mesopotamia)

### WEEK 11 📅 15m Story of Geometry: Part 1 - Lesson 23

*Circles as Instruments of Measure*

📎 Materials: String, Straight-Edge, and Shadow

### → RECAP

What did you read last time?

### → INTRO

Besides astronomy, in which fields of study or in what other endeavors do you think the circle could be used?

### → READ, NARRATE, & DISCUSS

Ch.6 p.54-57 (UK 48-51) "Apart from" - "Near Eastern geometry."

### WEEK 12 📅 15m Story of Geometry: Part 1 - Lesson 24

*Exam Week*

### → EXAMS

Answer question(s) related to course.

Questions will come from:

String, Straight-Edge, and Shadow

# Story of Geometry: Part 1

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## Term 3

### WEEK 1 15m Story of Geometry: Part 1 - Lesson 25

*Babylonians & Greeks*

Materials: String, Straight-Edge, and Shadow; Student Bible Atlas

#### → RECAP

What do you remember from last term?

#### → READ, NARRATE, & DISCUSS

Ch.7 p.59-60 (UK 54-55) "At the start" - "the Black Sea."

#### → LOCATE

Map Resource: Student Bible Atlas "Map 13: Babylon About 550 B.C." p.16

Look at the map of Babylon. Locate the Mediterranean and Black Seas (not labeled). Find Nineveh. What river runs by it? What Bible stories do you remember about Nineveh or King Nebuchadnezzar?

#### • IMPORTANT DATES

First Theoretical Geometry (c.6th century BC, Ionia)

Thales of Miletus (c.624 BC-c.548 BC)

### WEEK 2 15m Story of Geometry: Part 1 - Lesson 26

*From Practical to Theoretical Geometry*

Materials: String, Straight-Edge, and Shadow; Historical Atlas of the World

#### → RECAP

What did you read last time?

#### → INTRO

Map Resource: Historical Atlas of the World "Classical Greece and Athenian Empire About 450 B.C." p.7

Find the colony of Ionia. Look to the south to find the city of Miletus. This is where today's story takes place.

#### → READ, NARRATE, & DISCUSS

Ch.7 p.60-61 (UK 55-57) "So the 6th century" - "first scientists."

• Tell how the location of Miletus influenced its knowledge and development.

### WEEK 3 15m Story of Geometry: Part 1 - Lesson 27

*Thales: Founder of Geometry*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.7 p.61-62 (UK 57) "The Greeks had" - "founder of geometry."

### WEEK 4 15m Story of Geometry: Part 1 - Lesson 28

*Thales and Scientific Inquiry*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.8 p.63-64 (UK 58-59) "Thales, the father" - "the next season."

# Story of Geometry: Part 1

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## Term 3

### WEEK 5 15m Story of Geometry: Part 1 - Lesson 29

*Importance of Observation*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What was Thales studying in last week's lesson? What do you think he will do?

#### → READ, NARRATE, & DISCUSS

Ch.8 p.65-67 (UK 59-61) "Now he made" - "under your feet."

- Tell one of the stories of Thales.

### WEEK 6 15m Story of Geometry: Part 1 - Lesson 30

*Practical + Theoretical = New Ideas*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.8 p.67-68 (UK 61-62) "Nobody likes" - "theoretical geometry."

### WEEK 7 15m Story of Geometry: Part 1 - Lesson 31

*Magic & Mathematics*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.9 p.69-70 (UK 63-64) "In the Land" - "was a magician."

#### → DISCUSS

How do you think Thales calculated the height of the Great Pyramid? Find something tall outside (a tree, telephone pole, etc.). Can you devise a way you could figure out its height?

### WEEK 8 15m Story of Geometry: Part 1 - Lesson 32

*Proportion*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

Were you able to figure out how to calculate the height of the tall object you chose?

#### → READ, NARRATE, & DISCUSS

Ch.9 p.70-72 (UK 64-66) "To be sure" - "abstract proportion."

### WEEK 9 15m Story of Geometry: Part 1 - Lesson 33

*Shadows & Right Triangles*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

# Story of Geometry: Part 1

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## Term 3

### → READ, NARRATE, & DISCUSS

Ch.9 p.73-75 (UK 66-67) "Where others saw" - "height of the pyramid."

#### WEEK 10 15m Story of Geometry: Part 1 - Lesson 34

*Finding the Height of a Tall Object*

Materials: String, Straight-Edge, and Shadow

### → RECAP

What did you read last time?

### → INTRO

Today we can try Thales' feat ourselves.

### → READ, NARRATE, & DISCUSS

Ch.9 p.75-76 (UK 67-70) "Before you hear" - "height of the pyramid."

### → PRACTICE

Use shadows and proportions to calculate the height of a tall object.

- Explain how Thales calculated the height of the pyramid.

#### WEEK 11 15m Story of Geometry: Part 1 - Lesson 35

*The Abstract Right Triangle*

Materials: String, Straight-Edge, and Shadow

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

Ch.9 p.76-78 (UK 70-71) "When he asked" - "of reasoning."

#### WEEK 12 15m Story of Geometry: Part 1 - Lesson 36

*Exam Week*

### → EXAMS

Answer question(s) related to course.

Questions will come from:

String, Straight-Edge, and Shadow

# Story of Geometry: Part 1

## Examination

- Term 1** · String, Straight-Edge, and Shadow: Tell what you know about geometry in nature. OR Tell how design mimics nature.
- Term 2** · String, Straight-Edge, and Shadow: Tell how geometry is useful in architecture. OR Tell how humans learned to measure the seasons.
- Term 3** · String, Straight-Edge, and Shadow: Draw and explain how Thales calculated the height of the Great Pyramid.